

Qualification Pack



Solar Photovoltaic Entrepreneur

Electives: Rooftop/ Pumping/ Ground mounted- Civil/ Ground mounted-Electrical

QP Code: SGJ/Q0901

Version: 2.0

NSQF Level: 5

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SGJ/Q0901: Solar Photovoltaic Entrepreneur

Brief Job Description

Solar Photovoltaic Entrepreneur is an individual who ventures into Solar market to lead an enterprise. He/ She has the understanding of solar business models, market, technical knowledge of rooftop solar PV plants, along with components procurement and financing. Solar Photovoltaic Entrepreneur can prepare the feasibility study report and perform basic energy generation forecasting using simulation software and is also responsible for the managing the complete Solar PV project lifecycle.

Personal Attributes

The individual shall have an ability to take initiative, handle pressure and shall be motivated with good interpersonal and problem solving skills. The individual shall have leadership skills and organized with their work and act with integrity while performing multiple tasks for the customers with quality and timely deliverables.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N0111: Set up New Venture for Solar Photovoltaic System](#)
2. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)
3. [SGJ/N0106: Maintain Personal Health & Safety at project site](#)

Electives (mandatory to select at least one):

Elective 1: Rooftop

This Elective comprises 5 units which provides a detailed understanding of market research while preparing cost estimation for a rooftop solar photovoltaic plant, a detailed overview on preparing a site feasibility study report (Rooftop Solar), an overview of how to manage installation, operation & maintenance of solar rooftop power plant as well as Solar PV Project Cycle.

1. [SGJ/N0108: Carry out market research and prepare a cost estimate for a Rooftop Solar Photovoltaic plant](#)
2. [SGJ/N0109: Prepare a site feasibility study report](#)
3. [SGJ/N0148: Manage Installation, Operation & Maintenance of Solar Rooftop Power Plant](#)
4. [SGJ/N0110: Manage Solar PV Project Cycle](#)

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Elective 2: Pumping

This elective comprises 5 units which provides a detailed overview on developing solar PV pumping business, preparation of site feasibility for water pumping, a brief understanding on how to manage installation process, operation & maintenance of solar water pumps as well as solar PV water pumping project lifecycle

1. [SGJ/N0143: Develop Solar PV Pumping Business](#)
2. [SGJ/N1206: Prepare a site feasibility of Solar PV water pump](#)
3. [SGJ/N0405: Manage Installation, operation and maintenance of Solar water pumps](#)
4. [SGJ/N0404: Manage Solar PV water pumping Project lifecycle](#)

Elective 3: Ground mounted- Civil

This Elective comprises 3 units which provides a detailed understanding about Site Feasibility Study for Ground Mount Solar power plant, interpreting civil and structural design of solar PV power plant and Managing civil foundation and mounting structure for installation of solar PV power plant

1. [SGJ/N0125: Survey site for installation of ground mount solar PV power Plant](#)
2. [SGJ/N0301: Interpret civil and structural design of solar PV power plant](#)
3. [SGJ/N0403: Manage civil foundation and mounting structure for installation of solar PV power plant](#)

Elective 4: Ground mounted-Electrical

This Elective comprises 3 units which provides a detailed understanding about Electrical feasibility study for ground mount solar power plant, Interpreting Electrical drawing of ground mount solar plant and Managing AC and DC side of ground mount solar PV power plant.

1. [SGJ/N1207: Electrical feasibility study for ground mount solar power plant](#)
2. [SGJ/N0402: Interpret Electrical drawing of ground mount solar plant](#)
3. [SGJ/N0401: Manage AC and DC side of ground mount solar PV power plant](#)

Qualification Pack (QP) Parameters

Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Entrepreneur

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Country	India
NSQF Level	5
Credits	42
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2431.0501
Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) with NA of experience OR Pursuing 2nd year of UG (and continuous education) OR Completed 2nd year diploma after 12th with NA of experience OR Pursuing 2nd year of 2-year diploma after 12th with NA of experience OR 12th pass with 1 year Vocational Education & training (NTC or NAC or CITS) with NA of experience OR Completed 3 year diploma after 10th with 1 Year of experience relevant experience OR 12th grade Pass with 2 Years of experience relevant experience OR 10th grade pass with 4 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level (NSQF Level 4 and with minimum education as 8th Grade pass) with 3 Years of experience relevant experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	27/01/2025
NSQC Approval Date	27/01/2022
Version	2.0
Reference code on NQR	QG-05-ES-00528-2023-V1.1-SCGJ



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NQR Version	1
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Remarks:

Total 540 hours i.e. 18 Credits (Including 150 hours mandatory + 60 hours Employability Skills + 90 hours OJT + 240 hours of 1 elective)

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SGJ/N0111: Set up New Venture for Solar Photovoltaic System

Description

This unit provides an overview of Solar PV Sector with evolving business opportunities in terms of technology, finance & regulatory landscape of solar energy. This unit also provides a brief understanding about developing entrepreneurship skills for starting a new venture and managing it.

Scope

The scope covers the following :

- Introduction to Solar PV sector in India
- Set up new venture for Solar Photovoltaic

Elements and Performance Criteria

Introduction to Solar PV Sector in India

To be competent, the user/individual on the job must be able to:

- PC1.** discuss and provide overview of solar PV technology, rooftop and ground mounted solar sector in India
- PC2.** discuss various market research reports and industrial magazines present in the market
- PC3.** discuss different type of ground mounted and Rooftop Solar PV Power Plants and their working principles
- PC4.** discuss and provide an overview of grid interactive and off- grid Solar Sector in India
- PC5.** discuss type of off- grid Solar PV Power devices and their working principles
- PC6.** discuss various components of a solar PV system and their operating principles
- PC7.** discuss solar energy and power sector landscape in the country
- PC8.** discuss benefits of solar energy over conventional sources of energy
- PC9.** discuss the broader business, finance, technology & regulatory landscape of solar energy in the country
- PC10.** discuss how to analyse the basics of electrical concepts like voltage, current, power, energy, etc
- PC11.** discuss typical specifications, functioning, operating principle, maintenance requirements, handling procedures and warranties of different types of solar PV plant components
- PC12.** discuss how to interpret sample Electricity bill of a customer, notices and/or cautions
- PC13.** discuss how to identify various financial institutions and banks involved in lending for solar power projects and analyse their terms & conditions associated with loans
- PC14.** discuss emerging solar entrepreneurship landscape including success stories of various start-ups

Set up new venture for Solar Photovoltaic

To be competent, the user/individual on the job must be able to:

- PC15.** discuss the process for setting up a new venture including required registrations and availing low-cost financing from Financing Institutions (FIs) and Venture Capital (VC) firms

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- PC16.** discuss various evolving opportunities across solar solutions, services and market ecosystem
- PC17.** discuss how to identify the key ingredients of a business plan
- PC18.** discuss how to distinguish between fixed and working capital requirements
- PC19.** discuss the components of a loan application
- PC20.** discuss how to demonstrate leadership skills and effective resource management techniques
- PC21.** discuss the use of MS Word and Excel for preparing a business proposal
- PC22.** discuss to prepare a workable presentation for marketing and business development
- PC23.** discuss how to choose the right buyer in a given situation of market parameters
- PC24.** discuss how to identify the challenges and risks for new entrepreneurs
- PC25.** discuss various taxation and related compliances to set up and operate a solar business
- PC26.** discuss the key aspects to successfully run a solar business
- PC27.** discuss how to evolving Start- ups across solar Consultancy/O&M/Installation and EPC space, and diversifying solar portfolio is improving overall solar landscape
- PC28.** show how to the process for setting up a new venture including registrations and availing financing from FIs and VCs
- PC29.** show how to identify the key ingredients of a business plan
- PC30.** show the process of a loan application
- PC31.** show the application of tools including MS Word and Excel for preparing a business proposal
- PC32.** show how to prepare a workable presentation for marketing and business development
- PC33.** show how to choose the right buyer in a given situation of market parameters
- PC34.** show how to identify the challenges and risks for new entrepreneurs
- PC35.** show how to the importance of leadership skills and effective resource management techniques
- PC36.** show the key aspects to successfully run a solar business
- PC37.** show how evolving Start- ups across solar Consultancy/O&M/Installation and EPC, and diversifying solar portfolio is improving overall solar landscape
- PC38.** show how to adhere to various taxation and related compliances to set up and operate a solar business

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** government/corporate policies and guidelines on solar pv, solar rooftop
- KU2.** companys work safety policy
- KU3.** company's customer support policy.
- KU4.** companys documentation policy.
- KU5.** obtain authorization from specified field safety officer and supervisor.
- KU6.** companys different department.
- KU7.** definition of entrepreneurship from different perspectives

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- KU8.** outline the importance of entrepreneurship: enhances creativity and innovation, builds self confidence in people, serves as a tool for nation building, serves as the engine of growth for the nations economy.
- KU9.** explain the reasons why entrepreneurship should be developed in a country: reasons include: employment generation, increased national production and re-investing national resources
- KU10.** state the characteristics of an entrepreneur: characteristics of the entrepreneurs, risk taking, innovation and creativity, opportunity orientation
- KU11.** explain the challenges/problems facing small businesses like financing and access to markets, government policies and inadequate managerial skills
- KU12.** describe the procedure for registering a business by defining a business idea, source of business idea, programs/ procedure and available schemes.
- KU13.** state the process of starting a new enterprises process by mobilizing and reorganizing resources.
- KU14.** study of different pictorial expression of non-verbal communication and its analysis
- KU15.** time management concepts including discipline, punctuality, act in time on commitment and quality productive time
- KU16.** ability to shape and direct working/process methods according to self-defined criteria
- KU17.** empathize: comprehend other opinions points of views, and face them with understanding
- KU18.** learn ms word and ms excel: creating, organizing & formatting content, collaborating merge, insert, view, edit, track mode etc.
- KU19.** understand the fixed and capital working requirements for running a business
- KU20.** understand how to make a business plan

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare and maintain documentation
- GS2.** read vernacular/English language
- GS3.** read and understand manuals, health and safety instructions, memos, other company documents
- GS4.** read from different sources- books, screens in machines and signage
- GS5.** read various colour codes, as per standard electrical, mechanical and civil nomenclature
- GS6.** express statements or information clearly so that others can hear and understand
- GS7.** participate in and understand the main points of simple discussions
- GS8.** respond appropriately to any queries
- GS9.** communicate with employees.
- GS10.** define organization rule- based decision making process
- GS11.** take decision with systematic course of actions and/or response
- GS12.** plan and organize work schedule to meet deadlines
- GS13.** work constructively and collaboratively with others
- GS14.** prepare organization code of conduct

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- GS15.** manage relationships with customers with intent on satisfying its requirements for service delivery
- GS16.** recognize problems and search for solutions
- GS17.** choose best methods to complete assigned tasks
- GS18.** apply domain knowledge, observations and data to select course of action to perform tasks related to solar photovoltaic power plant.
- GS19.** critically evaluate information obtained from customers and workers to perform day to day activities
- GS20.** ask questions for better understanding
- GS21.** practice and acceptance of gender and its concepts
- GS22.** empathy across genders and towards PwD
- GS23.** communicate with team members and colleagues on the significance of greening of jobs
- GS24.** make timely decisions for efficient utilization of resources

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Solar PV Sector in India</i>	28	-	-	-
PC1. discuss and provide overview of solar PV technology, rooftop and ground mounted solar sector in India	2	-	-	-
PC2. discuss various market research reports and industrial magazines present in the market	2	-	-	-
PC3. discuss different type of ground mounted and Rooftop Solar PV Power Plants and their working principles	2	-	-	-
PC4. discuss and provide an overview of grid interactive and off- grid Solar Sector in India	2	-	-	-
PC5. discuss type of off- grid Solar PV Power devices and their working principles	2	-	-	-
PC6. discuss various components of a solar PV system and their operating principles	2	-	-	-
PC7. discuss solar energy and power sector landscape in the country	2	-	-	-
PC8. discuss benefits of solar energy over conventional sources of energy	2	-	-	-
PC9. discuss the broader business, finance, technology & regulatory landscape of solar energy in the country	2	-	-	-
PC10. discuss how to analyse the basics of electrical concepts like voltage, current, power, energy, etc	2	-	-	-
PC11. discuss typical specifications, functioning, operating principle, maintenance requirements, handling procedures and warranties of different types of solar PV plant components	2	-	-	-
PC12. discuss how to interpret sample Electricity bill of a customer, notices and/or cautions	2	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. discuss how to identify various financial institutions and banks involved in lending for solar power projects and analyse their terms & conditions associated with loans	2	-	-	-
PC14. discuss emerging solar entrepreneurship landscape including success stories of various start-ups	2	-	-	-
<i>Set up new venture for Solar Photovoltaic</i>	39	33	-	-
PC15. discuss the process for setting up a new venture including required registrations and availing low-cost financing from Financing Institutions (FIs) and Venture Capital (VC) firms	3	-	-	-
PC16. discuss various evolving opportunities across solar solutions, services and market ecosystem	3	-	-	-
PC17. discuss how to identify the key ingredients of a business plan	3	-	-	-
PC18. discuss how to distinguish between fixed and working capital requirements	3	-	-	-
PC19. discuss the components of a loan application	3	-	-	-
PC20. discuss how to demonstrate leadership skills and effective resource management techniques	3	-	-	-
PC21. discuss the use of MS Word and Excel for preparing a business proposal	3	-	-	-
PC22. discuss to prepare a workable presentation for marketing and business development	3	-	-	-
PC23. discuss how to choose the right buyer in a given situation of market parameters	3	-	-	-
PC24. discuss how to identify the challenges and risks for new entrepreneurs	3	-	-	-
PC25. discuss various taxation and related compliances to set up and operate a solar business	3	-	-	-
PC26. discuss the key aspects to successfully run a solar business	3	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. discuss how to evolving Start- ups across solar Consultancy/O&M/Installation and EPC space, and diversifying solar portfolio is improving overall solar landscape	3	-	-	-
PC28. show how to the process for setting up a new venture including registrations and availing financing from FIs and VCs	-	3	-	-
PC29. show how to identify the key ingredients of a business plan	-	3	-	-
PC30. show the process of a loan application	-	3	-	-
PC31. show the application of tools including MS Word and Excel for preparing a business proposal	-	3	-	-
PC32. show how to prepare a workable presentation for marketing and business development	-	3	-	-
PC33. show how to choose the right buyer in a given situation of market parameters	-	3	-	-
PC34. show how to identify the challenges and risks for new entrepreneurs	-	3	-	-
PC35. show how to the importance of leadership skills and effective resource management techniques	-	3	-	-
PC36. show the key aspects to successfully run a solar business	-	3	-	-
PC37. show how evolving Start- ups across solar Consultancy/O&M/Installation and EPC, and diversifying solar portfolio is improving overall solar landscape	-	3	-	-
PC38. show how to adhere to various taxation and related compliances to set up and operate a solar business	-	3	-	-
NOS Total	67	33	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0111
NOS Name	Set up New Venture for Solar Photovoltaic System
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Entrepreneur
NSQF Level	5
Credits	4
Version	3.0
Last Reviewed Date	NA
Next Review Date	27/01/2025
NSQC Clearance Date	27/01/2022

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

SGJ/N0106: Maintain Personal Health & Safety at project site

Description

This unit is about maintaining health, safety and hygiene at workplace.

Scope

The scope covers the following :

- Adopt safe practices at workplace
- Follow emergencies, rescue and first aid procedures
- Follow good housekeeping practices and infection control guidelines

Elements and Performance Criteria

Adopt safe practices at workplace

To be competent, the user/individual on the job must be able to:

- PC1.** explain the requirements for safe work area
- PC2.** identify and report any hazards, risks or breaches in site safety to the appropriate authority
- PC3.** follow recommended safe practices in handling physical, chemical, electrical and fire hazards and risk
- PC4.** use appropriate Personal Protective Equipment(PPE) for head, eye, hand, ear, face, body and fall protection specific to work condition
- PC5.** follow safe practices when working at height and in confined space
- PC6.** handle all required tools, tackles, materials and equipment safely
- PC7.** identify expiry dates, wear and tear issues of specified equipment and accordingly inform supervisor and undertake corrective measures
- PC8.** apply ergonomic principles wherever required
- PC9.** use safety signs, labels, charts and notices at workplace
- PC10.** identify work safety procedures and instructions for handling heavy components

Follow emergencies, rescue and first aid procedures

To be competent, the user/individual on the job must be able to:

- PC11.** follow emergency and evacuation procedures in case of accidents, fires and natural calamities
- PC12.** use appropriate fire extinguishers for different types of fire
- PC13.** administer first aid to victim in case of various medical emergencies including bleeding, burns, choking, electric shock, cardiac arrest, etc.
- PC14.** use correct method to move injured person during an emergency

Follow good housekeeping practices and infection control guidelines

To be competent, the user/individual on the job must be able to:

- PC15.** follow recommended personal hygiene, workplace hygiene and sanitation practices
- PC16.** clean and disinfect all material, tools and supplies before and after use

Qualification Pack

- PC17.** report immediately to concerned authorities regarding sign and symptoms of illness of self and other colleagues
- PC18.** follow processes specified for disposal of hazardous waste

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** importance of safety drills
- KU2.** importance of working in clean and safe environment
- KU3.** health and safety roles and responsibilities of relevant personnel within and outside the organization
- KU4.** reporting procedures in case of breaches or hazards for site safety, accidents and emergency situations
- KU5.** basic ergonomic principle
- KU6.** key internal and external source of health and safety information
- KU7.** meaning of hazards, risk and near miss
- KU8.** importance of Personal Protective Equipment required for specific job
- KU9.** forms and classification of hazardous substances
- KU10.** health effect associated with exposure to environmental pollution
- KU11.** housekeeping activities relevant to task
- KU12.** symptoms of infection like fever, cough, swelling and inflammation

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record data on waste disposal at workplace
- GS2.** complete statutory documents relevant to safety and hygiene
- GS3.** fill safety formats for near miss, unsafe condition
- GS4.** identify potential safety risk and report to appropriate authority
- GS5.** communicate and collaborate with others to incorporate sustainable practices
- GS6.** interpret general safety guidelines, labels, charts and signage

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Adopt safe practices at workplace</i>	13	19	-	-
PC1. explain the requirements for safe work area	2	-	-	-
PC2. identify and report any hazards, risks or breaches in site safety to the appropriate authority	2	3	-	-
PC3. follow recommended safe practices in handling physical, chemical, electrical and fire hazards and risk	1	2	-	-
PC4. use appropriate Personal Protective Equipment(PPE) for head, eye, hand, ear, face, body and fall protection specific to work condition	2	4	-	-
PC5. follow safe practices when working at height and in confined space	1	1	-	-
PC6. handle all required tools, tackles, materials and equipment safely	1	2	-	-
PC7. identify expiry dates, wear and tear issues of specified equipment and accordingly inform supervisor and undertake corrective measures	1	2	-	-
PC8. apply ergonomic principles wherever required	1	2	-	-
PC9. use safety signs, labels, charts and notices at workplace	1	1	-	-
PC10. identify work safety procedures and instructions for handling heavy components	1	2	-	-
<i>Follow emergencies, rescue and first aid procedures</i>	4	4	-	-
PC11. follow emergency and evacuation procedures in case of accidents, fires and natural calamities	1	1	-	-
PC12. use appropriate fire extinguishers for different types of fire	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. administer first aid to victim in case of various medical emergencies including bleeding, burns, choking, electric shock, cardiac arrest, etc.	1	1	-	-
PC14. use correct method to move injured person during an emergency	1	1	-	-
<i>Follow good housekeeping practices and infection control guidelines</i>	4	6	-	-
PC15. follow recommended personal hygiene, workplace hygiene and sanitation practices	1	1	-	-
PC16. clean and disinfect all material, tools and supplies before and after use	1	1	-	-
PC17. report immediately to concerned authorities regarding sign and symptoms of illness of self and other colleagues	1	2	-	-
PC18. follow processes specified for disposal of hazardous waste	1	2	-	-
NOS Total	21	29	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0106
NOS Name	Maintain Personal Health & Safety at project site
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Health & Safety , Installation, Solar Panel Installation Technician
NSQF Level	4
Credits	1
Version	4.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

SGJ/N0108: Carry out market research and prepare a cost estimate for a Rooftop Solar Photovoltaic plant

Description

This unit is about assessing the cost of system installation and identification of various business models

Scope

The scope covers the following :

- Market Research and cost estimation of rooftop solar project

Elements and Performance Criteria

Market Research and cost estimation of rooftop solar project

To be competent, the user/individual on the job must be able to:

- PC1.** discuss how to assess the market trends and evaluate the business prospect for rooftop solar
- PC2.** discuss how to identify market opportunities and potential customers
- PC3.** discuss how to select the right quality of solar module by identifying the key technical parameters in data specification sheets
- PC4.** discuss how to select the right quality of inverter by identifying the key technical parameters in data specification sheets
- PC5.** discuss how to select the right quality of mounting structure by identifying the key technical parameters in data specification sheets
- PC6.** discuss how to select the right quality of battery by identifying the key technical parameters in data specification sheets
- PC7.** discuss how to identify market price of different components of solar PV system
- PC8.** discuss how to prepare cost estimate for a rooftop solar project
- PC9.** discuss how to prepare a cost benefit analysis for a rooftop solar PV plant including Levelised Cost of Energy (LCOE), Payback, Internal Rate of Return (IRR) etc
- PC10.** discuss how to identify the policy, regulations and procedures for solar rooftop sector in the local market
- PC11.** discuss how to identify and select the appropriate business models in solar rooftop sector
- PC12.** show how to identify different segments of market looking for Solar PV solutions
- PC13.** show how to prepare list of companies on material supply for rooftop solar business
- PC14.** show how to select the balance of system by identifying the key technical and financial parameters
- PC15.** show how to undertake the cost benefit analysis for setting up of rooftop solar PV power plant including performing the calculation of Levelised Cost of Energy (LCOE), payback, Internal Rate of Return (IRR), etc
- PC16.** show how to prepare a proposal for setting up of rooftop solar PV power plant
- PC17.** show how to analyse policy, regulations and procedures for solar rooftop sector
- PC18.** show how to identify and select the appropriate business models in solar rooftop sector

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company's installation policy
- KU2.** company's customer support policy
- KU3.** company's documentation policy
- KU4.** document information using appropriate corporate forms
- KU5.** obtain authorization from specified field safety officer and supervisor
- KU6.** definition of the general terminology used by the industry
- KU7.** knowledge of all the technical parameters and interpretation of specification sheets of different commercially available components
- KU8.** efficiency, cost and typical specifications, functioning and operating principle of different types of solar photovoltaic plants, commercially available pv modules, inverters, charge controllers, battery, mounting structures, cables, junction boxes and other components
- KU9.** prepare costing and cost benefit analysis for project including lcoe, payback, irr etc.
- KU10.** project budgeting
- KU11.** business models for solar rooftop sector like capex, opex, boot, etc.
- KU12.** policy, regulations and procedures for installing a rooftop solar pv power plant
- KU13.** availing available central and /or state subsidy for various consumer segments

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare and maintain documentation.
- GS2.** read vernacular/english language
- GS3.** read and understand manuals, health and safety instructions, memos, other company documents
- GS4.** read from different sources- books, screens in machines and signage.
- GS5.** read various colour codes, as per standard electrical, mechanical and civil nomenclature
- GS6.** express statements or information clearly so that others can hear and understand
- GS7.** participate in and understand the main points of simple discussions
- GS8.** respond appropriately to any queries
- GS9.** communicate with employees
- GS10.** define organization rule- based decision making process
- GS11.** take decision with systematic course of actions and/or response
- GS12.** plan and organize work schedule to meet deadlines.
- GS13.** work constructively and collaboratively with others
- GS14.** prepare organization code of conduct.
- GS15.** manage relationships with customers with intent on satisfying its requirements for service delivery.
- GS16.** recognize problems and search for solutions.



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- GS17.** choose best methods to complete assigned tasks.
- GS18.** apply domain knowledge, observations and data to select course of action to perform tasks related to solar photovoltaic power plant.
- GS19.** critically evaluate information obtained from customers and workers to perform day to day activities.
- GS20.** ask questions for better understanding.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Market Research and cost estimation of rooftop solar project</i>	29	21	-	-
PC1. discuss how to assess the market trends and evaluate the business prospect for rooftop solar	2	-	-	-
PC2. discuss how to identify market opportunities and potential customers	2	-	-	-
PC3. discuss how to select the right quality of solar module by identifying the key technical parameters in data specification sheets	2	-	-	-
PC4. discuss how to select the right quality of inverter by identifying the key technical parameters in data specification sheets	2	-	-	-
PC5. discuss how to select the right quality of mounting structure by identifying the key technical parameters in data specification sheets	3	-	-	-
PC6. discuss how to select the right quality of battery by identifying the key technical parameters in data specification sheets	3	-	-	-
PC7. discuss how to identify market price of different components of solar PV system	3	-	-	-
PC8. discuss how to prepare cost estimate for a rooftop solar project	3	-	-	-
PC9. discuss how to prepare a cost benefit analysis for a rooftop solar PV plant including Levelised Cost of Energy (LCOE), Payback, Internal Rate of Return (IRR) etc	3	-	-	-
PC10. discuss how to identify the policy, regulations and procedures for solar rooftop sector in the local market	3	-	-	-
PC11. discuss how to identify and select the appropriate business models in solar rooftop sector	3	-	-	-
PC12. show how to identify different segments of market looking for Solar PV solutions	-	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. show how to prepare list of companies on material supply for rooftop solar business	-	3	-	-
PC14. show how to select the balance of system by identifying the key technical and financial parameters	-	3	-	-
PC15. show how to undertake the cost benefit analysis for setting up of rooftop solar PV power plant including performing the calculation of Levelised Cost of Energy (LCOE), payback, Internal Rate of Return (IRR), etc	-	3	-	-
PC16. show how to prepare a proposal for setting up of rooftop solar PV power plant	-	3	-	-
PC17. show how to analyse policy, regulations and procedures for solar rooftop sector	-	3	-	-
PC18. show how to identify and select the appropriate business models in solar rooftop sector	-	3	-	-
NOS Total	29	21	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0108
NOS Name	Carry out market research and prepare a cost estimate for a Rooftop Solar Photovoltaic plant
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Entrepreneur
NSQF Level	5
Credits	2
Version	3.0
Last Reviewed Date	NA
Next Review Date	27/01/2025
NSQC Clearance Date	27/01/2022

Qualification Pack

SGJ/N0109: Prepare a site feasibility study report

Description

This unit is about the key steps involved in developing a site feasibility report for rooftop solar PV system along with assessing client requirement

Scope

The scope covers the following :

- Introduction to Solar PV Sector in India
- Prepare a site Feasibility Study Report

Elements and Performance Criteria

Introduction to Solar PV Sector in India

To be competent, the user/individual on the job must be able to:

- PC1.** discuss and provide an overview of solar PV technology
- PC2.** discuss and provide an overview of ground mounted and rooftop solar sector
- PC3.** discuss types of ground mounted PV Power Plants and their working principles
- PC4.** discuss various types of Rooftop Solar PV Power Plant and working principles
- PC5.** discuss solar PV water pumping system technology and market
- PC6.** discuss and provide an overview of National and State specific Solar PV water pumping related policies (e.g. PM KUSUM) and regulations
- PC7.** discuss and provide an overview of available subsidies and financing options of solar water pumping system
- PC8.** show the function of any solar PV system including solar water pumping system
- PC9.** demonstrate know how of installation of solar water pump.
- PC10.** show how financing for solar water pumping system can be arranged and how subsidies can be availed

Prepare a site Feasibility Study Report

To be competent, the user/individual on the job must be able to:

- PC11.** discuss how to identify optimum location for installations
- PC12.** discuss different types of Roofs and their structural load profile
- PC13.** discuss how to assess the site level pre-requisites for solar panel installation on both ground and roof
- PC14.** discuss how to analyse type of mounting to be constructed and place of mounting as per client requirement
- PC15.** discuss how to check for any shading obstacles
- PC16.** discuss how to prepare a site map of the location where installation has to be carried out
- PC17.** discuss how to assess the load to be run on solar PV power plant and prepare a load profile
- PC18.** discuss how to estimate the capacity of solar PV power plant

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- PC19.** discuss how to decide on battery backup as per grid availability, loads and client expectation
- PC20.** discuss how to assess or obtain the site specific major parameters of solar resource data like Global Horizontal Irradiance (GHI), Diffused Normal Irradiance (DNI), Temperature and Wind
- PC21.** discuss how to perform shading analysis using simulation tools
- PC22.** discuss how to identify the risks associated with the specific solar project
- PC23.** discuss how to prepare a site Feasibility Study Report using specialized software like PV*SOL®, PVsyst, SketchUp etc
- PC24.** show how to perform simple calculations to illustrate the fundamental concepts of power and energy
- PC25.** show how the movement of sun affects the performance of the solar power plant
- PC26.** show how to obtain site specific resource data like GHI, DNI etc
- PC27.** show how to estimate the energy generated from the solar PV power plant using solar design software like PV*SOL®, PVsyst, etc..
- PC28.** show how to prepare a site feasibility report

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** companys policies on: incentives, personnel management
- KU2.** companys code of conduct
- KU3.** importance of individuals role in the work flow
- KU4.** companys documentation policy
- KU5.** companys installation policy
- KU6.** companys customer support policy
- KU7.** simple calculations to derive the power and energy received from solar radiation in a given area
- KU8.** solar resource assessment including direct normal irradiation, diffuse horizontal irradiation, global horizontal irradiation and albedo
- KU9.** ground based measurement and satellite derived data
- KU10.** building orientation
- KU11.** types of roofs and suggestive mounting structure for that specific roof
- KU12.** basic concepts of Trigonometry and coordinate geometry
- KU13.** effect on array output of current and voltage based on series / parallel connections of modules, tilt angle, orientation and shading
- KU14.** effect of shading on the system using tools like Solar path finder and softwares like PV*SOL, PVsyst, etc.
- KU15.** cabling route and length of cable required
- KU16.** different types of tracking systems
- KU17.** simulation software
- KU18.** risks associated with the solar project
- KU19.** synchronization with diesel generators

Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read warnings, instructions and other text material on product labels, components etc.
- GS2.** express statements for information clearly so that customer can hear and understand
- GS3.** follow code of conduct
- GS4.** manage relationships with customers with intent on satisfying its requirements for service delivery
- GS5.** interact with client to analyse client exact requirement
- GS6.** interact with workers in order to coordinate work processes
- GS7.** participate in and understand main points of simple discussions with customer
- GS8.** respond properly to any query of the customer

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Solar PV Sector in India</i>	21	15	-	-
PC1. discuss and provide an overview of solar PV technology	3	-	-	-
PC2. discuss and provide an overview of ground mounted and rooftop solar sector	3	-	-	-
PC3. discuss types of ground mounted PV Power Plants and their working principles	3	-	-	-
PC4. discuss various types of Rooftop Solar PV Power Plant and working principles	3	-	-	-
PC5. discuss solar PV water pumping system technology and market	3	-	-	-
PC6. discuss and provide an overview of National and State specific Solar PV water pumping related policies (e.g. PM KUSUM) and regulations	3	-	-	-
PC7. discuss and provide an overview of available subsidies and financing options of solar water pumping system	3	-	-	-
PC8. show the function of any solar PV system including solar water pumping system	-	5	-	-
PC9. demonstrate know how of installation of solar water pump.	-	5	-	-
PC10. show how financing for solar water pumping system can be arranged and how subsidies can be availed	-	5	-	-
<i>Prepare a site Feasibility Study Report</i>	39	25	-	-
PC11. discuss how to identify optimum location for installations	3	-	-	-
PC12. discuss different types of Roofs and their structural load profile	3	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. discuss how to assess the site level pre-requisites for solar panel installation on both ground and roof	3	-	-	-
PC14. discuss how to analyse type of mounting to be constructed and place of mounting as per client requirement	3	-	-	-
PC15. discuss how to check for any shading obstacles	3	-	-	-
PC16. discuss how to prepare a site map of the location where installation has to be carried out	3	-	-	-
PC17. discuss how to assess the load to be run on solar PV power plant and prepare a load profile	3	-	-	-
PC18. discuss how to estimate the capacity of solar PV power plant	3	-	-	-
PC19. discuss how to decide on battery backup as per grid availability, loads and client expectation	3	-	-	-
PC20. discuss how to assess or obtain the site specific major parameters of solar resource data like Global Horizontal Irradiance (GHI), Diffused Normal Irradiance (DNI), Temperature and Wind	3	-	-	-
PC21. discuss how to perform shading analysis using simulation tools	3	-	-	-
PC22. discuss how to identify the risks associated with the specific solar project	3	-	-	-
PC23. discuss how to prepare a site Feasibility Study Report using specialized software like PV*SOL®, PVsyst, SketchUp etc	3	-	-	-
PC24. show how to perform simple calculations to illustrate the fundamental concepts of power and energy	-	5	-	-
PC25. show how the movement of sun affects the performance of the solar power plant	-	5	-	-
PC26. show how to obtain site specific resource data like GHI, DNI etc	-	5	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. show how to estimate the energy generated from the solar PV power plant using solar design software like PV*SOL®, PVsyst, etc..	-	5	-	-
PC28. show how to prepare a site feasibility report	-	5	-	-
NOS Total	60	40	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0109
NOS Name	Prepare a site feasibility study report
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Solar Site Survey
NSQF Level	5
Credits	2
Version	4.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

SGJ/N0148: Manage Installation, Operation & Maintenance of Solar Rooftop Power Plant

Description

This unit is about managing installation, operation and maintenance of Solar Rooftop power plant

Scope

The scope covers the following :

- Installation of Solar PV rooftop plant
- Operation and maintenance of Solar PV rooftop plant

Elements and Performance Criteria

Installation of Solar PV rooftop plant

To be competent, the user/individual on the job must be able to:

- PC1.** explain how to oversee casting of civil foundation on roof
- PC2.** ensure installation tools availability on site
- PC3.** oversee the process of installing mounting structure
- PC4.** discuss installation of foundation for ACDB, Inverter etc
- PC5.** explain how to install the Energy meter
- PC6.** explain how to install the AC isolator
- PC7.** discuss how to install the Earthing and Lighting arrestor on site
- PC8.** explain use of electrical installation tools
- PC9.** explain the process of testing and commissioning of equipment in DC and AC side of the plant
- PC10.** show how to different components and tools used in rooftop solar power plant installation
- PC11.** show how to oversee installation of solar rooftop solar power plant
- PC12.** show how to analyse the process of overseeing the installation of various mounting structures
- PC13.** show how to oversee the testing and commissioning of equipment in DC and AC side of the solar rooftop plant

Operation and maintenance of Solar PV rooftop plant

To be competent, the user/individual on the job must be able to:

- PC14.** explain how to use tools & equipment for Operation and maintenance (O&M) of solar rooftop plant
- PC15.** explain how to check the integrity and working condition of all connections, fuses and circuit breakers
- PC16.** ensure working condition of the junction boxes/combiner boxes
- PC17.** ensure the continuity of cables and wires are maintained
- PC18.** ensure proper electrical connections up to the inverter input

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- PC19.** discuss how to verify the connections, cables and junction boxes as per the design/ working drawings
- PC20.** ensure escalation of issues of identified fault to concerned person for timely resolution
- PC21.** explain how to verify the earthing and lightening protection systems as per the as-built drawings and report in case of any discrepancies
- PC22.** ensure proper cleaning of modules as per schedule and standard procedure and remove any shadowing objects
- PC23.** discuss how to implement approaches for O&M including scheduled/preventive and unscheduled or corrective maintenance
- PC24.** ensure that the system is well maintained and is performing at an optimum level
- PC25.** identify and incorporate learnings from the best practices and successful case studies on performing O&M
- PC26.** show how to handle tools and equipment for O&M available at site
- PC27.** show how to check the integrity and working condition of all connections, fuses, circuit breakers and all other components of the rooftop solar power plant
- PC28.** show how to verify the connections, cables and junction boxes as per the design/ working drawings
- PC29.** show how to ensure that the system is well maintained and is performing at an optimum level
- PC30.** show how to implement approaches for O&M including scheduled/preventive and unscheduled or corrective maintenance
- PC31.** show how to interpret a successful case study on operation and maintenance of solar rooftop power plant
- PC32.** show how to best practices on O&M as employed by the industry

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** manufacturer's warranty policy
- KU2.** tools & tackles required for installation
- KU3.** effect on array output of current and voltage based on series / parallel connections of modules, tilt angle, orientation and shading
- KU4.** mechanical and electrical features necessary for the long life of the pv system under a wide range of operating conditions
- KU5.** type of mounting structure required depending upon the type of roof
- KU6.** type of footings and fixtures required depending upon the type of roof
- KU7.** cabling route and estimate the length of cable required
- KU8.** dos and don'ts of material handling and storage
- KU9.** relevant standards and regulations for resource utilization
- KU10.** occupational health and safety (OHS) standards and associated risks when working at particular site
- KU11.** relevant codes, standards, guidelines and regulations for grid interactive solar rooftop installation and commissioning



Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and understand equipment manuals, health and safety instructions, memos, other concerned documents
- GS2.** take decision with systematic course of actions and/or response
- GS3.** analyze critical points in day to day tasks and identify control measures to solve the issue
- GS4.** support the team in scheduling tasks

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Installation of Solar PV rooftop plant</i>	19	16	-	-
PC1. explain how to oversee casting of civil foundation on roof	2	-	-	-
PC2. ensure installation tools availability on site	2	-	-	-
PC3. oversee the process of installing mounting structure	2	-	-	-
PC4. discuss installation of foundation for ACDB, Inverter etc	2	-	-	-
PC5. explain how to install the Energy meter	3	-	-	-
PC6. explain how to install the AC isolator	2	-	-	-
PC7. discuss how to install the Earthing and Lighting arrestor on site	2	-	-	-
PC8. explain use of electrical installation tools	2	-	-	-
PC9. explain the process of testing and commissioning of equipment in DC and AC side of the plant	2	-	-	-
PC10. show how to different components and tools used in rooftop solar power plant installation	-	4	-	-
PC11. show how to oversee installation of solar rooftop solar power plant	-	4	-	-
PC12. show how to analyse the process of overseeing the installation of various mounting structures	-	4	-	-
PC13. show how to oversee the testing and commissioning of equipment in DC and AC side of the solar rooftop plant	-	4	-	-
<i>Operation and maintenance of Solar PV rooftop plant</i>	35	30	-	-
PC14. explain how to use tools & equipment for Operation and maintenance (O&M) of solar rooftop plant	2	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. explain how to check the integrity and working condition of all connections, fuses and circuit breakers	3	-	-	-
PC16. ensure working condition of the junction boxes/combiner boxes	3	-	-	-
PC17. ensure the continuity of cables and wires are maintained	3	-	-	-
PC18. ensure proper electrical connections up to the inverter input	3	-	-	-
PC19. discuss how to verify the connections, cables and junction boxes as per the design/ working drawings	3	-	-	-
PC20. ensure escalation of issues of identified fault to concerned person for timely resolution	3	-	-	-
PC21. explain how to verify the earthing and lightening protection systems as per the as-built drawings and report in case of any discrepancies	3	-	-	-
PC22. ensure proper cleaning of modules as per schedule and standard procedure and remove any shadowing objects	3	-	-	-
PC23. discuss how to implement approaches for O&M including scheduled/preventive and unscheduled or corrective maintenance	3	-	-	-
PC24. ensure that the system is well maintained and is performing at an optimum level	3	-	-	-
PC25. identify and incorporate learnings from the best practices and successful case studies on performing O&M	3	-	-	-
PC26. show how to handle tools and equipment for O&M available at site	-	5	-	-
PC27. show how to check the integrity and working condition of all connections, fuses, circuit breakers and all other components of the rooftop solar power plant	-	5	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC28. show how to verify the connections, cables and junction boxes as per the design/ working drawings	-	4	-	-
PC29. show how to ensure that the system is well maintained and is performing at an optimum level	-	4	-	-
PC30. show how to implement approaches for O&M including scheduled/preventive and unscheduled or corrective maintenance	-	4	-	-
PC31. show how to interpret a successful case study on operation and maintenance of solar rooftop power plant	-	4	-	-
PC32. show how to best practices on O&M as employed by the industry	-	4	-	-
NOS Total	54	46	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0148
NOS Name	Manage Installation, Operation & Maintenance of Solar Rooftop Power Plant
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Installation, Operation and Maintenance
NSQF Level	5
Credits	2
Version	2.0
Last Reviewed Date	NA
Next Review Date	27/01/2025
NSQC Clearance Date	27/01/2022

Qualification Pack

SGJ/N0110: Manage Solar PV Project Cycle

Description

This unit is about the implementation schedule and maintenance of rooftop solar PV project for effective functioning to achieve the specified output

Scope

The scope covers the following :

- Manage solar PV project lifecycle

Elements and Performance Criteria

Manage Solar PV Project lifecycle

To be competent, the user/individual on the job must be able to:

- PC1.** discuss how to read and interpret the single line diagram, civil / mechanical drawings and electrical drawings
- PC2.** discuss how to read and interpret the bill of material.
- PC3.** discuss how to calculate the lifecycle cost of a rooftop solar project
- PC4.** discuss how to identify and mitigate various risks associated with the project
- PC5.** discuss how to ensure the solar PV system and structure meets the local government and regulatory requirements
- PC6.** discuss how to prepare action plan and coordinate the implementation of rooftop solar project
- PC7.** discuss how to identify the maintenance activity required for a rooftop solar PV power plant component
- PC8.** discuss how to prepare a preventive maintenance schedule
- PC9.** discuss how to ensure proper cleaning of solar panels periodically
- PC10.** discuss how to ensure regular inspection of the solar PV system to identify and rectify the faults
- PC11.** show how to read and interpret the single line diagram, civil / mechanical drawings and electrical drawings
- PC12.** show how to read and interpret the bill of material
- PC13.** show how to calculate the lifecycle cost of a rooftop solar project
- PC14.** show how to prepare action plan and coordinate the implementation of rooftop solar project
- PC15.** show how to identify and perform the maintenance activity required for a rooftop solar PV power plant
- PC16.** show how to prepare a preventive maintenance schedule
- PC17.** show how to ensure proper cleaning of solar panels periodically and perform regular inspection of the solar PV system to identify and rectify the faults

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** legislative, organization, site requirements and procedures.
- KU2.** the environmental requirements
- KU3.** work in varying weather conditions
- KU4.** single Line Diagram, Civil / Mechanical drawings, Electrical drawings and how to read a Bill of Material for a rooftop solar PV power plant
- KU5.** knowledge of Excel and Solar simulation softwares like PV*SOL, PVsyst, etc.
- KU6.** debt and equity financing options
- KU7.** work in varying weather conditions
- KU8.** different risks associated with the project like cost over-run, project completion risk, company risk related to key personnel and technical ability to execute on plans, environmental risk, financial risk, market risk, operational risk, technology risk and regulatory risk
- KU9.** preparation of an activity implementation plan for a rooftop solar project
- KU10.** work in varying weather conditions
- KU11.** preparation of preventive maintenance schedule for different components
- KU12.** tools involved in maintenance
- KU13.** work in varying weather conditions
- KU14.** safety precautions to be taken while handling different solar pv power plant components
- KU15.** knowledge of waste segregation categories
- KU16.** knowledge of required process for waste disposal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare and maintain proper documentation.
- GS2.** ability to read from different sources- books screens in machines and signage.
- GS3.** understand the various color codes, as per standard electrical, mechanical and civil nomenclature.
- GS4.** to clearly communicate installation and design instructions to team.
- GS5.** to clearly communicate customers requirements.
- GS6.** respond appropriately to any queries.
- GS7.** to communicate the constraints and quality requirements to team.
- GS8.** take decision with systematic course of actions and/or response.
- GS9.** planning and organization of work to meet deadlines.
- GS10.** follow code of conduct.
- GS11.** manage relationships with customers with intent on satisfying its requirements for quality delivery.
- GS12.** recognize problems and search for solutions.
- GS13.** choose best methods to complete assigned tasks.



Qualification Pack

- GS14.** approach relevant authority when required.
- GS15.** apply domain knowledge, observations and data to select course of action to perform tasks related to solar photovoltaic systems.
- GS16.** critically evaluate information obtained from customers and workers to perform day to day activities.
- GS17.** ask questions for better understanding.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Manage Solar PV Project lifecycle</i>	29	21	-	-
PC1. discuss how to read and interpret the single line diagram, civil / mechanical drawings and electrical drawings	2	-	-	-
PC2. discuss how to read and interpret the bill of material.	3	-	-	-
PC3. discuss how to calculate the lifecycle cost of a rooftop solar project	3	-	-	-
PC4. discuss how to identify and mitigate various risks associated with the project	3	-	-	-
PC5. discuss how to ensure the solar PV system and structure meets the local government and regulatory requirements	3	-	-	-
PC6. discuss how to prepare action plan and coordinate the implementation of rooftop solar project	3	-	-	-
PC7. discuss how to identify the maintenance activity required for a rooftop solar PV power plant component	3	-	-	-
PC8. discuss how to prepare a preventive maintenance schedule	3	-	-	-
PC9. discuss how to ensure proper cleaning of solar panels periodically	3	-	-	-
PC10. discuss how to ensure regular inspection of the solar PV system to identify and rectify the faults	3	-	-	-
PC11. show how to read and interpret the single line diagram, civil / mechanical drawings and electrical drawings	-	3	-	-
PC12. show how to read and interpret the bill of material	-	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. show how to calculate the lifecycle cost of a rooftop solar project	-	3	-	-
PC14. show how to prepare action plan and coordinate the implementation of rooftop solar project	-	3	-	-
PC15. show how to identify and perform the maintenance activity required for a rooftop solar PV power plant	-	3	-	-
PC16. show how to prepare a preventive maintenance schedule	-	3	-	-
PC17. show how to ensure proper cleaning of solar panels periodically and perform regular inspection of the solar PV system to identify and rectify the faults	-	3	-	-
NOS Total	29	21	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0110
NOS Name	Manage Solar PV Project Cycle
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Solar Project management
NSQF Level	5
Credits	2
Version	3.0
Last Reviewed Date	NA
Next Review Date	27/01/2025
NSQC Clearance Date	27/01/2022

Qualification Pack

SGJ/N0143: Develop Solar PV Pumping Business

Description

This unit is about developing solar PV pumping business while assessing the market trends in solar pumping business, the quality of solar components as well as the cost of solar water pump

Scope

The scope covers the following :

- Market Research and cost estimation of solar PV water pump business

Elements and Performance Criteria

Market Research and cost estimation of solar PV water pump business

To be competent, the user/individual on the job must be able to:

- PC1.** discuss how to assess the market trends and evaluate the business prospect for solar water pumping
- PC2.** discuss how to identify market opportunities and potential customers
- PC3.** discuss how to select the right quality of solar module by identifying the key technical parameters in data specification sheets
- PC4.** discuss how to select the right quality of material and mounting structure by identifying the key technical parameters in data specification sheets
- PC5.** discuss how to select the right quality of battery by identifying the key technical parameters in data specification sheets
- PC6.** discuss how to identify market price of different components of solar water pump
- PC7.** discuss how to prepare cost estimate for a solar water pump
- PC8.** discuss how to prepare a cost benefit analysis for a solar water pumping including Levelised Cost of Energy (LCOE), Payback, Internal Rate of Return (IRR) etc.
- PC9.** discuss how to identify the policy, regulations and procedures for deployment of solar water pumping system
- PC10.** discuss how to identify and select the appropriate business models in solar water pumping
- PC11.** show how to identify different segments of market looking for solar water pumps
- PC12.** show how to prepare list of companies on material supply for solar water pumping
- PC13.** show how to select the balance of system by identifying the key technical and financial parameters
- PC14.** show how to prepare the cost benefit analysis for setting up of solar water pump for agriculture usages including the calculation of Levelised Cost of Energy (LCOE), payback, Internal Rate of Return (IRR), etc
- PC15.** show how to prepare a proposal for setting up of solar water pumping plant
- PC16.** show how to analyse policy, regulations and procedures for solar water pumping
- PC17.** show how to identify and select the appropriate business models in solar rooftop sector

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** government policies and subsidies for solar PV pumps
- KU2.** the market trends and evaluate the business prospect for solar water pumping
- KU3.** market opportunities and potential customers
- KU4.** select the right quality of solar module by identifying the key technical parameters in data specification sheets
- KU5.** the right quality of material and mounting structure by identifying the key technical parameters in data specification sheets
- KU6.** the right quality of battery by identifying the key technical parameters in data specification sheets
- KU7.** market price of different components of solar water pump
- KU8.** cost estimate for a solar water pump.
- KU9.** cost benefit analysis for a solar water pumping
- KU10.** the policy, regulations and procedures for deployment of solar water pumping system.
- KU11.** appropriate business models in solar water pumping

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare and maintain documentation
- GS2.** read vernacular/English language
- GS3.** read and understand manuals, health and safety instructions, memos, other company documents
- GS4.** read from different sources- books, screens in machines and signage
- GS5.** read various colour codes, as per standard electrical, mechanical and civil nomenclature
- GS6.** express statements or information clearly so that others can hear and understand
- GS7.** participate in and understand the main points of simple discussions
- GS8.** respond appropriately to any queries
- GS9.** communicate with employees
- GS10.** define organisation rule- based decision making process
- GS11.** take decision with systematic course of actions and/or response
- GS12.** plan and organize work schedule to meet deadlines
- GS13.** work constructively and collaboratively with others
- GS14.** prepare organisation code of conduct
- GS15.** manage relationships with customers with intent on satisfying its requirements for service delivery
- GS16.** recognize problems and provide solutions using a range of cognitive and practical skills
- GS17.** apply knowledge of facts, principles and processes to select the right course of action to perform tasks

Qualification Pack

- GS18.** use common sense and make judgments on day to day basis
- GS19.** use reasoning skills to identify and resolve basic problems
- GS20.** use intuition to detect any potential problems which could arise during operations
- GS21.** use acquired knowledge of the process for identifying and handling issues
- GS22.** complete statutory documents relevant to safety and hygiene
- GS23.** • communicate with industries and customers to understand and analyze various strategies,
• demands, and limitations in the market

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Market Research and cost estimation of solar PV water pump business</i>	29	21	-	-
PC1. discuss how to assess the market trends and evaluate the business prospect for solar water pumping	2	-	-	-
PC2. discuss how to identify market opportunities and potential customers	3	-	-	-
PC3. discuss how to select the right quality of solar module by identifying the key technical parameters in data specification sheets	3	-	-	-
PC4. discuss how to select the right quality of material and mounting structure by identifying the key technical parameters in data specification sheets	3	-	-	-
PC5. discuss how to select the right quality of battery by identifying the key technical parameters in data specification sheets	3	-	-	-
PC6. discuss how to identify market price of different components of solar water pump	3	-	-	-
PC7. discuss how to prepare cost estimate for a solar water pump	3	-	-	-
PC8. discuss how to prepare a cost benefit analysis for a solar water pumping including Levelised Cost of Energy (LCOE), Payback, Internal Rate of Return (IRR) etc.	3	-	-	-
PC9. discuss how to identify the policy, regulations and procedures for deployment of solar water pumping system	3	-	-	-
PC10. discuss how to identify and select the appropriate business models in solar water pumping	3	-	-	-
PC11. show how to identify different segments of market looking for solar water pumps	-	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. show how to prepare list of companies on material supply for solar water pumping	-	3	-	-
PC13. show how to select the balance of system by identifying the key technical and financial parameters	-	3	-	-
PC14. show how to prepare the cost benefit analysis for setting up of solar water pump for agriculture usages including the calculation of Levelised Cost of Energy (LCOE), payback, Internal Rate of Return (IRR), etc	-	3	-	-
PC15. show how to prepare a proposal for setting up of solar water pumping plant	-	3	-	-
PC16. show how to analyse policy, regulations and procedures for solar water pumping	-	3	-	-
PC17. show how to identify and select the appropriate business models in solar rooftop sector	-	3	-	-
NOS Total	29	21	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0143
NOS Name	Develop Solar PV Pumping Business
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Business Development
NSQF Level	5
Credits	2
Version	3.0
Last Reviewed Date	NA
Next Review Date	27/01/2025
NSQC Clearance Date	27/01/2022

Qualification Pack

SGJ/N1206: Prepare a site feasibility of Solar PV water pump

Description

This unit is about preparation of site feasibility report for installation of solar PV water pump

Scope

The scope covers the following :

- Site Feasibility Study of solar PV water pump

Elements and Performance Criteria

Site Feasibility Study of solar PV water pump

To be competent, the user/individual on the job must be able to:

- PC1.** discuss how to assess site for solar PV pump installation
- PC2.** discuss how to collect data of local weather conditions
- PC3.** discuss how to conduct study for drinking and irrigation water requirement of the user/s and it is being presently met
- PC4.** discuss how to analyse the depth of the water/ ground water source and the quantity/recharge period
- PC5.** discuss how to assess the soil test report
- PC6.** discuss how to create relevant solar pumping model of appropriate capacity and specifications
- PC7.** discuss how to prepare the cost benefit analysis of the solar pump and make a presentation to the customer
- PC8.** discuss how to identify manufacturers and suppliers through various sources who can meet the requirements
- PC9.** show how to assess site for solar water pump installation and prepare feasibility report
- PC10.** show how to analyse the depth of the water/ ground water source and the quantity/recharge period
- PC11.** show how to use site survey tools at sites
- PC12.** show how to prepare the feasibility report

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** basic concepts of Trigonometry and coordinate geometry
- KU2.** type of DC/AC pumping systems usage and their operating characteristics
- KU3.** water levels, availability of water and recharging frequency of water

Generic Skills (GS)



Qualification Pack

User/individual on the job needs to know how to:

GS1. organize raw materials and packaging materials required for site survey

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Site Feasibility Study of solar PV water pump</i>	30	20	-	-
PC1. discuss how to assess site for solar PV pump installation	3	-	-	-
PC2. discuss how to collect data of local weather conditions	4	-	-	-
PC3. discuss how to conduct study for drinking and irrigation water requirement of the user/s and it is being presently met	3	-	-	-
PC4. discuss how to analyse the depth of the water/ ground water source and the quantity/recharge period	3	-	-	-
PC5. discuss how to assess the soil test report	4	-	-	-
PC6. discuss how to create relevant solar pumping model of appropriate capacity and specifications	4	-	-	-
PC7. discuss how to prepare the cost benefit analysis of the solar pump and make a presentation to the customer	4	-	-	-
PC8. discuss how to identify manufacturers and suppliers through various sources who can meet the requirements	5	-	-	-
PC9. show how to assess site for solar water pump installation and prepare feasibility report	-	5	-	-
PC10. show how to analyse the depth of the water/ ground water source and the quantity/recharge period	-	5	-	-
PC11. show how to use site survey tools at sites	-	5	-	-
PC12. show how to prepare the feasibility report	-	5	-	-
NOS Total	30	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N1206
NOS Name	Prepare a site feasibility of Solar PV water pump
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Site survey
NSQF Level	5
Credits	2
Version	2.0
Last Reviewed Date	NA
Next Review Date	27/01/2025
NSQC Clearance Date	27/01/2022

Qualification Pack

SGJ/N0405: Manage Installation, operation and maintenance of Solar water pumps

Description

This unit is about to ensure installation, operation and maintenance of Solar water pumps

Scope

The scope covers the following :

- Installation of Solar PV water pump
- Operation and maintenance of Solar PV water pump

Elements and Performance Criteria

Installation of Solar PV water pump

To be competent, the user/individual on the job must be able to:

- PC1.** identify and use tools for solar pump installation
- PC2.** decide specification of the solar pumping system including for DC and AC range of pumps suitable for Surface, Borewell and Open well applications
- PC3.** determine the head of the location of pump installation
- PC4.** oversee the preparation of Civil foundation for water pumping system
- PC5.** ensure installation of mounting structure
- PC6.** ensure electrical installation tools availability at site
- PC7.** ensure installation of foundation for ACDB and Inverter
- PC8.** ensure installation of Earthing and Lighting arrestor on site
- PC9.** ensure installation of control electronics, motor mounting structure, pumps and other components of solar water pumping system
- PC10.** demonstrate assembling of solar water pumping along with AC isolator and Energy meter
- PC11.** demonstrate drilling of bore and placing of pump into the bore
- PC12.** discuss how to the properly place the pump into the bore
- PC13.** show how to identify and use tools for solar pump installation
- PC14.** show how to perform inspection and testing of equipment
- PC15.** show how to analyse specifications of the solar pumping system
- PC16.** show how to analyse the water depth
- PC17.** show how to analyse the testing and commissioning procedure of solar water pumping system
- PC18.** show how to oversee the installation of the solar pumping systems
- PC19.** show how to various key technical aspects of a Pump controller and identify the process for trouble shooting of a Solar PV Pumping system

Operation and maintenance of Solar PV water pump

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC20.** discuss how to check the integrity and working condition of all components, fuses and circuit breakers
- PC21.** ensure working condition of the junction boxes, MPPT, controller and all other key electrical component
- PC22.** ensure the continuity of cables, wires and other electrical components
- PC23.** ensure troubleshooting of identified faults in controller and pumps
- PC24.** ensure proper cleaning of solar modules as per schedule
- PC25.** prepare records for key system parameters required in preventive and predictive maintenance
- PC26.** show how to troubleshoot failure of integration of solar pump and module
- PC27.** show how to perform the visual inspection of system
- PC28.** show how to perform the system operation
- PC29.** show how to perform cleaning of modules as per schedule and undertake standard procedure for ensuring optimal performance of the solar pumping system
- PC30.** show how to analyse how to perform optimal system operation by monitoring and recording key system parameters and timely implementing preventive and predictive maintenance procedures

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** to identify and understand the functions of key components of solar water pump
- KU2.** to identify and understand the functions of key components of solar water pump
- KU3.** displacement and centrifugal pump
- KU4.** water demand of an area
- KU5.** to identify and understand the functions of key components of solar water pump
- KU6.** ground water levels and its recharging frequency
- KU7.**
 - efficiency, cost and typical specifications, functioning and operating principle of different
 - types of solar photovoltaic plants, commercially available PV modules, inverters, charge
 - controllers, battery, mounting structures, cables, junction boxes and other components

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- GS2.** identify skilled technician for pump installation and solar integration and post sales maintenance
- GS3.** lead a team of group of technicians
- GS4.** plan and organize service work to meet deadlines
- GS5.** use reasoning skills to identify and resolve basic problems
- GS6.** manage relationships with customers with intent on satisfying its requirements for service delivery



Qualification Pack

GS7. plan to utilize time and equipment's effectively

GS8. provide water as a service delivery

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Installation of Solar PV water pump</i>	34	28	-	-
PC1. identify and use tools for solar pump installation	2	-	-	-
PC2. decide specification of the solar pumping system including for DC and AC range of pumps suitable for Surface, Borewell and Open well applications	3	-	-	-
PC3. determine the head of the location of pump installation	2	-	-	-
PC4. oversee the preparation of Civil foundation for water pumping system	3	-	-	-
PC5. ensure installation of mounting structure	3	-	-	-
PC6. ensure electrical installation tools availability at site	3	-	-	-
PC7. ensure installation of foundation for ACDB and Inverter	3	-	-	-
PC8. ensure installation of Earthing and Lighting arrestor on site	3	-	-	-
PC9. ensure installation of control electronics, motor mounting structure, pumps and other components of solar water pumping system	3	-	-	-
PC10. demonstrate assembling of solar water pumping along with AC isolator and Energy meter	3	-	-	-
PC11. demonstrate drilling of bore and placing of pump into the bore	3	-	-	-
PC12. discuss how to the properly place the pump into the bore	3	-	-	-
PC13. show how to identify and use tools for solar pump installation	-	4	-	-
PC14. show how to perform inspection and testing of equipment	-	4	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. show how to analyse specifications of the solar pumping system	-	4	-	-
PC16. show how to analyse the water depth	-	4	-	-
PC17. show how to analyse the testing and commissioning procedure of solar water pumping system	-	4	-	-
PC18. show how to oversee the installation of the solar pumping systems	-	4	-	-
PC19. show how to various key technical aspects of a Pump controller and identify the process for trouble shooting of a Solar PV Pumping system	-	4	-	-
<i>Operation and maintenance of Solar PV water pump</i>	18	20	-	-
PC20. discuss how to check the integrity and working condition of all components, fuses and circuit breakers	3	-	-	-
PC21. ensure working condition of the junction boxes, MPPT, controller and all other key electrical component	3	-	-	-
PC22. ensure the continuity of cables, wires and other electrical components	3	-	-	-
PC23. ensure troubleshooting of identified faults in controller and pumps	3	-	-	-
PC24. ensure proper cleaning of solar modules as per schedule	3	-	-	-
PC25. prepare records for key system parameters required in preventive and predictive maintenance	3	-	-	-
PC26. show how to troubleshoot failure of integration of solar pump and module	-	4	-	-
PC27. show how to perform the visual inspection of system	-	4	-	-
PC28. show how to perform the system operation	-	4	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. show how to perform cleaning of modules as per schedule and undertake standard procedure for ensuring optimal performance of the solar pumping system	-	4	-	-
PC30. show how to analyse how to perform optimal system operation by monitoring and recording key system parameters and timely implementing preventive and predictive maintenance procedures	-	4	-	-
NOS Total	52	48	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0405
NOS Name	Manage Installation, operation and maintenance of Solar water pumps
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Management
NSQF Level	5
Credits	2
Version	2.0
Last Reviewed Date	NA
Next Review Date	27/01/2025
NSQC Clearance Date	27/01/2022

Qualification Pack

SGJ/N0404: Manage Solar PV water pumping Project lifecycle

Description

This unit is about managing solar PV water pumping project lifecycle

Scope

The scope covers the following :

- Solar PV water pump Project lifecycle

Elements and Performance Criteria

Solar PV water pump Project lifecycle

To be competent, the user/individual on the job must be able to:

- PC1.** discuss how to read and interpret the single line diagram, civil / mechanical drawings and electrical drawings
- PC2.** discuss how to read and interpret the bill of material
- PC3.** discuss how to calculate the lifecycle cost of a solar water pump
- PC4.** discuss how to identify and mitigate various risks associated with the project
- PC5.** discuss how to ensure the solar water pump and structure meets the local government and regulatory requirements
- PC6.** discuss how to prepare action plan and coordinate the implementation of solar water pumping project
- PC7.** discuss how to identify the maintenance activity required for a solar water pump component
- PC8.** discuss how to prepare a preventive maintenance schedule.
- PC9.** discuss how to ensure proper cleaning of solar panels periodically
- PC10.** discuss how to ensure regular inspection of the solar water pump to identify and rectify the faults.
- PC11.** show how to read and interpret the single line diagram, civil / mechanical drawings and electrical drawings
- PC12.** show how to read and interpret the bill of material
- PC13.** show how to calculate the lifecycle cost of a solar water pumping project
- PC14.** show how to prepare action plan and coordinate the implementation of solar water pump project
- PC15.** show how to identify and perform the maintenance activity required for a solar water pumping
- PC16.** show how to prepare a preventive maintenance schedule
- PC17.** show how to ensure proper cleaning of solar panels periodically and perform regular inspection of the solar water pump to identify and rectify the faults

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** bill of material required for solar water pump
- KU2.** risk associated with solar water pumping business
- KU3.** government and regulatory requirement for solar pumping business
- KU4.** various schemes promoting solar pumping business
- KU5.** to identify and understand the functions of key components of solar water pump
- KU6.** to identify and understand the functions of key components of solar water pump
- KU7.** displacement and centrifugal pump

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** organize and document various folders required in project management
- GS2.** communicate clearly among sub-ordinates respecting gender and disabilities
- GS3.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- GS4.** identify skilled technician for pump installation and solar integration and post sales maintenance
- GS5.** lead a team of group of technicians
- GS6.** plan and organize service work to meet deadlines
- GS7.** use reasoning skills to identify and resolve basic problems
- GS8.** manage relationships with customers with intent on satisfying its requirements for service delivery
- GS9.** plan to utilize time and equipment's effectively
- GS10.** provide water as a service delivery

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Solar PV water pump Project lifecycle</i>	29	21	-	-
PC1. discuss how to read and interpret the single line diagram, civil / mechanical drawings and electrical drawings	2	-	-	-
PC2. discuss how to read and interpret the bill of material	3	-	-	-
PC3. discuss how to calculate the lifecycle cost of a solar water pump	3	-	-	-
PC4. discuss how to identify and mitigate various risks associated with the project	3	-	-	-
PC5. discuss how to ensure the solar water pump and structure meets the local government and regulatory requirements	3	-	-	-
PC6. discuss how to prepare action plan and coordinate the implementation of solar water pumping project	3	-	-	-
PC7. discuss how to identify the maintenance activity required for a solar water pump component	3	-	-	-
PC8. discuss how to prepare a preventive maintenance schedule.	3	-	-	-
PC9. discuss how to ensure proper cleaning of solar panels periodically	3	-	-	-
PC10. discuss how to ensure regular inspection of the solar water pump to identify and rectify the faults.	3	-	-	-
PC11. show how to read and interpret the single line diagram, civil / mechanical drawings and electrical drawings	-	3	-	-
PC12. show how to read and interpret the bill of material	-	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. show how to calculate the lifecycle cost of a solar water pumping project	-	3	-	-
PC14. show how to prepare action plan and coordinate the implementation of solar water pump project	-	3	-	-
PC15. show how to identify and perform the maintenance activity required for a solar water pumping	-	3	-	-
PC16. show how to prepare a preventive maintenance schedule	-	3	-	-
PC17. show how to ensure proper cleaning of solar panels periodically and perform regular inspection of the solar water pump to identify and rectify the faults	-	3	-	-
NOS Total	29	21	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0404
NOS Name	Manage Solar PV water pumping Project lifecycle
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Management
NSQF Level	5
Credits	2
Version	2.0
Last Reviewed Date	NA
Next Review Date	27/01/2025
NSQC Clearance Date	27/01/2022

Qualification Pack

SGJ/N0125: Survey site for installation of ground mount solar PV power Plant

Description

This unit is about activities to be carried out under site survey for installation of solar PV power plant

Scope

The scope covers the following :

- Site Feasibility Study for Ground Mount Solar power plant

Elements and Performance Criteria

Site Feasibility Study for Ground Mount Solar power plant

To be competent, the user/individual on the job must be able to:

- PC1.** discuss how to assess irradiation and other geological data
- PC2.** discuss how to assess the daily, monthly and annual solar resource data including GHI, DNI, Albedo etc. for site to evaluate the potential for solar energy generation at the site
- PC3.** discuss how to collect data of local weather conditions such as temperature range, flooding, wind speed, humidity, pollution levels, snow and other climatic conditions for assessment of its impact on solar energy generation
- PC4.** show how to retrieve irradiation data from site
- PC5.** show how to analyse daily, monthly and annual solar resource data including GHI, DNI, Albedo etc. for site to evaluate the potential for solar energy generation at the site
- PC6.** show how to analyse the soil test report
- PC7.** show how to use site survey tools like, hand held irradiation meter, compass, Total station
- PC8.** discuss how to conduct the soil testing and contour mapping
- PC9.**
 - discuss how to conduct the detailed survey plan of the land proposed for installation of solar power plant with
 - elevations and topography contour mapping etc
- PC10.** discuss how to assess physical site accessibility
- PC11.** discuss how to identify the policy, regulations and procedures for ground mounted solar sector
- PC12.** discuss how to assess transmission line & power evacuation

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** government/corporate policies and guidelines on: workplace safety, identification and mitigation of safety hazards, work procedures and SOPs
- KU2.** document information using appropriate corporate forms
- KU3.** authorization formality for specified field safety and supervision

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- KU4.** legislative, organization, site requirements and procedures
- KU5.** basics on solar energy system and power generation
- KU6.** basics of solar resource assessment including terms like DNI, DHI, GHI and albedo and their interpretation
- KU7.** typical specifications, functioning, operating principle and efficiencies of different types of solar photovoltaic power plants, commercially available PV modules, inverters, charge controllers, battery, mounting structures, cables, junction boxes and other components
- KU8.** the different types of soils and its appropriateness for installing a solar PV power plant
- KU9.** site surveying methods and evaluation parameters
- KU10.** shading analysis and its importance and effect on the solar PV power plant design and effect of tilt angle orientation and shading on array output
- KU11.** ground based measurement and satellite derived data for solar resource and its variation with respect to energy generation
- KU12.** different types of tools used for land measurement
- KU13.** mechanical and electrical features necessary for the long life of the PV power plant under a wide range of operating conditions
- KU14.** simple calculations to derive the power and energy received from solar radiation in a given area
- KU15.** various parameters associated with local power loads/ grid dispatch
- KU16.** documentation parameters of site survey report

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare documentation as per relevant industry standards
- GS2.** present information in a logical and organized way
- GS3.** read English language
- GS4.** read and understand manuals, health and safety instructions, memos, other company documents
- GS5.** read and interpret data from various sources
- GS6.** express statements or information clearly so that others can hear and understand
- GS7.** participate in and understand the main points of simple discussions
- GS8.** communicate with peers, superiors and sub-ordinates
- GS9.** follow organization rule-based decision making process
- GS10.** take decision with systematic course of actions and/or response
- GS11.** planning and organization of work to meet deadlines
- GS12.** work constructively and collaboratively with others
- GS13.** support the superiors in scheduling tasks
- GS14.** follow organisation code of conduct
- GS15.** manage relationships with customers with intent on satisfying its requirements for service delivery
- GS16.** generate solutions to specific problems for a wide range of activities

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- GS17.** choose best methods to complete assigned tasks
- GS18.** apply wide range of factual and theoretical knowledge to select the right course of action to perform tasks related to solar photovoltaic power plant
- GS19.** use reasoning skills to identify and resolve basic problems
- GS20.** use intuition to detect any potential problems which could arise during operations
- GS21.** use acquired knowledge of the process for identifying and handling issues

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Site Feasibility Study for Ground Mount Solar power plant</i>	30	20	-	-
PC1. discuss how to assess irradiation and other geological data	3	-	-	-
PC2. discuss how to assess the daily, monthly and annual solar resource data including GHI, DNI, Albedo etc. for site to evaluate the potential for solar energy generation at the site	3	-	-	-
PC3. discuss how to collect data of local weather conditions such as temperature range, flooding, wind speed, humidity, pollution levels, snow and other climatic conditions for assessment of its impact on solar energy generation	4	-	-	-
PC4. show how to retrieve irradiation data from site	-	5	-	-
PC5. show how to analyse daily, monthly and annual solar resource data including GHI, DNI, Albedo etc. for site to evaluate the potential for solar energy generation at the site	-	5	-	-
PC6. show how to analyse the soil test report	-	5	-	-
PC7. show how to use site survey tools like, hand held irradiation meter, compass, Total station	-	5	-	-
PC8. discuss how to conduct the soil testing and contour mapping	4	-	-	-
PC9. - discuss how to conduct the detailed survey plan of the land proposed for installation of solar power plant with - elevations and topography contour mapping etc	4	-	-	-
PC10. discuss how to assess physical site accessibility	4	-	-	-
PC11. discuss how to identify the policy, regulations and procedures for ground mounted solar sector	4	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. discuss how to assess transmission line & power evacuation	4	-	-	-
NOS Total	30	20	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0125
NOS Name	Survey site for installation of ground mount solar PV power Plant
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Solar site survey
NSQF Level	5
Credits	3
Version	3.0
Last Reviewed Date	NA
Next Review Date	27/01/2025
NSQC Clearance Date	27/01/2022

Qualification Pack

SGJ/N0301: Interpret civil and structural design of solar PV power plant

Description

This unit is about interpreting civil and structural design of solar PV power plant

Scope

The scope covers the following :

- Interpretation of design and preparation of Civil foundation
- interpretation of layout, mounting structure and allied civil work design in ground mount solar plant

Elements and Performance Criteria

Interpretation of design and preparation of Civil foundation

To be competent, the user/individual on the job must be able to:

- PC1.** discuss how to prepare project development plan for civil foundation
- PC2.** discuss how to assess soil characteristics in accordance to mounting structure design
- PC3.** illustrate different types of foundation suitable for module mounting structures, inverters, transformers etc. based on the soil test report
- PC4.** discuss how to prepare structural design report
- PC5.** discuss how to ensure preparation of layout for cable trenching, Electrical control panel, Transformer, VCB etc
- PC6.** discuss how to prepare pile foundation, column post and other civil foundation
- PC7.** discuss how to design of the pile foundation as per the soil assessment report
- PC8.** discuss how to assess design of the civil/ structural allied works of the solar PV power plant including compound wall/entry gate internal plant roads walkways between different rows of modules water distribution network water drainage system, etc.
- PC9.** discuss the importance of soil test report for designing the Mounting structure.
- PC10.** discuss civil/ structural drawings for the ground mount solar power plant
- PC11.** discuss the reinforcement material for civil construction of various foundation
- PC12.** show how to read and interpret the layout drawing
- PC13.** show how to interpreted the Pile foundation and other civil foundation drawing.
- PC14.** show how to analyse the soil test report
- PC15.** show how to analyse the available weather data on site
- PC16.** show list out the tools required for installation of civil foundation and Mounting structure
- PC17.** show via the video and pics of tools and machinery used for construction of civil and mechanical works at site

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** legislative, organizational, site requirements and procedures
- KU2.** site survey and soil test reports for evaluation purposes
- KU3.** type of foundations of various components depending on the different types of soils and its appropriateness for installing a solar PV power plant
- KU4.** types of loads for structural load analysis
- KU5.** types of module mounting structures, its selection and design based on type of roof/ soil test report
- KU6.** types of allied civil/ structural works in a solar PV power plant
- KU7.** legislative requirements and organizations procedures for waste management and disposal
- KU8.** any relevant structural design software

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** analyze critical points in day to day tasks and identify control measures to solve the issue
- GS2.** plan and organize work to meet deadlines
- GS3.** work constructively and collaboratively with others
- GS4.** support the team in scheduling tasks
- GS5.** follow organizational rule-based decision making process
- GS6.** recognize problems and provide solutions

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Interpretation of design and preparation of Civil foundation</i>	32	18	-	-
PC1. discuss how to prepare project development plan for civil foundation	2	-	-	-
PC2. discuss how to assess soil characteristics in accordance to mounting structure design	2	-	-	-
PC3. illustrate different types of foundation suitable for module mounting structures, inverters, transformers etc. based on the soil test report	3	-	-	-
PC4. discuss how to prepare structural design report	3	-	-	-
PC5. discuss how to ensure preparation of layout for cable trenching, Electrical control panel, Transformer, VCB etc	3	-	-	-
PC6. discuss how to prepare pile foundation, column post and other civil foundation	3	-	-	-
PC7. discuss how to design of the pile foundation as per the soil assessment report	3	-	-	-
PC8. discuss how to assess design of the civil/ structural allied works of the solar PV power plant including compound wall/entry gate internal plant roads walkways between different rows of modules water distribution network water drainage system, etc.	4	-	-	-
PC9. discuss the importance of soil test report for designing the Mounting structure.	3	-	-	-
PC10. discuss civil/ structural drawings for the ground mount solar power plant	3	-	-	-
PC11. discuss the reinforcement material for civil construction of various foundation	3	-	-	-
PC12. show how to read and interpret the layout drawing	-	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. show how to interpreted the Pile foundation and other civil foundation drawing.	-	3	-	-
PC14. show how to analyse the soil test report	-	3	-	-
PC15. show how to analyse the available weather data on site	-	3	-	-
PC16. show list out the tools required for installation of civil foundation and Mounting structure	-	3	-	-
PC17. show via the video and pics of tools and machinery used for construction of civil and mechanical works at site	-	3	-	-
NOS Total	32	18	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0301
NOS Name	Interpret civil and structural design of solar PV power plant
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Engineering and Design
NSQF Level	5
Credits	2
Version	2.0
Last Reviewed Date	NA
Next Review Date	27/01/2025
NSQC Clearance Date	27/01/2022

Qualification Pack

SGJ/N0403: Manage civil foundation and mounting structure for installation of solar PV power plant

Description

This unit is about managing civil foundation and mounting structure for installation of solar PV power plant

Scope

The scope covers the following :

- Managing ground mounted solar Civil Construction work

Elements and Performance Criteria

Managing ground mounted solar Civil Construction work

To be competent, the user/individual on the job must be able to:

- PC1.** discuss how to prepare a project development plan
- PC2.** discuss the planning for construction of compounds wall or fencing, access of internal road, drainage
- PC3.** discuss how to develop the overall project constructability approach for cost effective implementation
- PC4.** discuss how to supervise and manage schedule for installation of module structures and modules, installation of inverters, transformers, earthing piles, DC/ AC power protection devices, lightning arresters, substation as per the grid codes and regulatory provisions
- PC5.** discuss how to plan, schedule, and monitor all material deliveries to avoid project delays
- PC6.** discuss how to manage schedule for each of the construction activity i.e. construction of internal roads, construction of foundations for mounting module support structures, combiner boxes, inverters, transformers, and substation etc
- PC7.**
 - discuss how to do successful completion of the projects by identifying innovative ways to maximize
 - Profit
- PC8.** discuss how to manage project billing cycle to ensure timely submission and payment of invoices
- PC9.** discuss how to analyze the bids to ensure complete and accurate quotes for scopes of work
- PC10.** discuss how to negotiate with subcontractors
- PC11.** discuss how to coordinate with the customers on a regular basis and provide them with the updates
- PC12.** discuss how to communicate to the customer and other key team members the contractual, financial, cost accounting, scheduling, technical and other matters related to the project
- PC13.** discuss periodic visits to the site(s) to review the progress at the ground
- PC14.** show how to prepare the Project development plan

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- PC15.** show how to prepare the project construction plan and to provide an overview to supervise and manage schedule for installation of module structures and modules, installation of inverters, transformers, earthing piles, DC/ AC power protection devices, lightning arresters, substation as per the grid codes and regulatory provisions
- PC16.** show how to oversee the implementation of schedule for each of the construction activity i.e. construction of internal roads, construction of foundations for mounting module support structures, combiner boxes, inverters, transformers, and substation etc
- PC17.** show how to effectively manage project billing cycle to ensure timely submission and payment of invoices
- PC18.** discuss the case studies of quality monitoring during the construction
- PC19.** show how to perform the handover process of particular work
- PC20.** show how to analyze the bids to ensure complete and accurate quotes for various scope of work

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** material and tools required for civil foundation construction work
- KU2.** coordinate with the customers on a regular basis
- KU3.** negotiate with subcontractors
- KU4.** communicate to the customer and other key team members
- KU5.** periodic visits to the site(s) to review the progress at the ground
- KU6.** effectively manage project

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** decide to choose make of material based on project size and budget
- GS2.** identify skill technician for civil works
- GS3.** effective communication with the customer and other key team members related to the project
- GS4.** coordination with the customers on a regular basis

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Managing ground mounted solar Civil Construction work</i>	32	18	-	-
PC1. discuss how to prepare a project development plan	2	-	-	-
PC2. discuss the planning for construction of compounds wall or fencing, access of internal road, drainage	2	-	-	-
PC3. discuss how to develop the overall project constructability approach for cost effective implementation	2	-	-	-
PC4. discuss how to supervise and manage schedule for installation of module structures and modules, installation of inverters, transformers, earthing piles, DC/ AC power protection devices, lightning arresters, substation as per the grid codes and regulatory provisions	3	-	-	-
PC5. discuss how to plan, schedule, and monitor all material deliveries to avoid project delays	2	-	-	-
PC6. discuss how to manage schedule for each of the construction activity i.e. construction of internal roads, construction of foundations for mounting module support structures, combiner boxes, inverters, transformers, and substation etc	2	-	-	-
PC7. • discuss how to do successful completion of the projects by identifying innovative ways to maximize • Profit	2	-	-	-
PC8. discuss how to manage project billing cycle to ensure timely submission and payment of invoices	2	-	-	-
PC9. discuss how to analyze the bids to ensure complete and accurate quotes for scopes of work	2	-	-	-
PC10. discuss how to negotiate with subcontractors	2	-	-	-
PC11. discuss how to coordinate with the customers on a regular basis and provide them with the updates	2	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. discuss how to communicate to the customer and other key team members the contractual, financial, cost accounting, scheduling, technical and other matters related to the project	3	-	-	-
PC13. discuss periodic visits to the site(s) to review the progress at the ground	3	-	-	-
PC14. show how to prepare the Project development plan	-	3	-	-
PC15. show how to prepare the project construction plan and to provide an overview to supervise and manage schedule for installation of module structures and modules, installation of inverters, transformers, earthing piles, DC/ AC power protection devices, lightning arresters, substation as per the grid codes and regulatory provisions	-	3	-	-
PC16. show how to oversee the implementation of schedule for each of the construction activity i.e. construction of internal roads, construction of foundations for mounting module support structures, combiner boxes, inverters, transformers, and substation etc	-	3	-	-
PC17. show how to effectively manage project billing cycle to ensure timely submission and payment of invoices	-	3	-	-
PC18. discuss the case studies of quality monitoring during the construction	3	-	-	-
PC19. show how to perform the handover process of particular work	-	3	-	-
PC20. show how to analyze the bids to ensure complete and accurate quotes for various scope of work	-	3	-	-
NOS Total	32	18	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0403
NOS Name	Manage civil foundation and mounting structure for installation of solar PV power plant
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Management
NSQF Level	5
Credits	3
Version	2.0
Last Reviewed Date	NA
Next Review Date	27/01/2025
NSQC Clearance Date	27/01/2022

Qualification Pack

SGJ/N1207: Electrical feasibility study for ground mount solar power plant

Description

This unit is about the key steps involved in developing a site electrical feasibility report for ground mounted solar plant.

Scope

The scope covers the following :

- Electrical Feasibility Study for Ground Mount Solar power plant

Elements and Performance Criteria

Electrical Feasibility Study for Ground Mount Solar power plant

To be competent, the user/individual on the job must be able to:

- PC1.** discuss how to assess site electrical feasibility
- PC2.** discuss how to assess cable root
- PC3.** discuss how to identify inverter location
- PC4.** discuss how to identify the transformer and other HT equipment location
- PC5.** discuss how to assess grid availability for power evacuation including nearest substation and transmission line capacity as well as distance from project site
- PC6.** discuss how to assess the evacuation voltage level i.e. 230 V/440 V/11 KV /33 KV66 kV
- PC7.** show how to identify optimum location for the ground mount project
- PC8.** show how to assess the site level pre-requisites for solar panel installation
- PC9.** show how to decide on the type of mounting to be constructed and place of mounting as per site condition based on soil test report and other relevant factor
- PC10.** show how to obtain the site-specific major parameters of solar resource data like GHI, DNI, Temperature and Wind
- PC11.** show how to estimate the energy generated from the solar PV power plant using solar design software like PV*SOL®, PVsyst, etc
- PC12.** show the risks associated with the specific solar project and mitigate those
- PC13.** show how to support a site Feasibility Study Report using specialized software like PV*SOL®, PVsyst, etc

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** government/corporate policies and guidelines on: workplace safety, identification and mitigation of safety hazards, work procedures and SOPs
- KU2.** authorization formality for specified field safety and supervision
- KU3.** legislative, organizational, site requirements and procedures for electrical connections in solar power plant installation

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- KU4.** typical specifications, functioning, operating principle and efficiencies of different types of solar photovoltaic power plants, commercially available PV modules, inverters, charge controllers, battery, mounting structures, cables, junction boxes and other components
- KU5.** site surveying methods and evaluation parameters
- KU6.** shading analysis and its importance and effect on the solar PV power plant design and effect of tilt angle orientation and shading on array output
- KU7.** ground based measurement and satellite derived data for solar resource and its variation with respect to energy generation
- KU8.** mechanical and electrical features necessary for the long life of the PV power plant under a wide range of operating conditions
- KU9.** various parameters associated with electrical safety and grid infrastructure requirements
- KU10.** documentation parameters of electrical feasibility and site survey report

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare documentation and reports as per relevant industry standards
- GS2.** present information in a logical and organized way
- GS3.** read and interpret data from various sources
- GS4.** express statements or information clearly so that others can hear and understand
- GS5.** communicate with peers, superiors and sub-ordinates
- GS6.** prepare documentation and reports as per relevant industry standards
- GS7.** take decision with systematic course of actions and/or response
- GS8.** work constructively and collaboratively with others
- GS9.** generate solutions to specific problems for a wide range of activities
- GS10.** choose best methods to complete assigned tasks
- GS11.** apply wide range of factual and theoretical knowledge to select the right course of action to perform tasks related to solar photovoltaic power plant
- GS12.** use reasoning skills to identify and resolve basic problems
- GS13.** use intuition to detect any potential problems which could arise during operations
- GS14.** use acquired knowledge of the process for identifying and handling issues

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Electrical Feasibility Study for Ground Mount Solar power plant</i>	18	32	-	-
PC1. discuss how to assess site electrical feasibility	3	-	-	-
PC2. discuss how to assess cable route	3	-	-	-
PC3. discuss how to identify inverter location	3	-	-	-
PC4. discuss how to identify the transformer and other HT equipment location	3	-	-	-
PC5. discuss how to assess grid availability for power evacuation including nearest substation and transmission line capacity as well as distance from project site	3	-	-	-
PC6. discuss how to assess the evacuation voltage level i.e. 230 V/440 V/11 KV /33 KV66 kV	3	-	-	-
PC7. show how to identify optimum location for the ground mount project	-	5	-	-
PC8. show how to assess the site level pre-requisites for solar panel installation	-	4	-	-
PC9. show how to decide on the type of mounting to be constructed and place of mounting as per site condition based on soil test report and other relevant factor	-	5	-	-
PC10. show how to obtain the site-specific major parameters of solar resource data like GHI, DNI, Temperature and Wind	-	5	-	-
PC11. show how to estimate the energy generated from the solar PV power plant using solar design software like PV*SOL®, PVsyst, etc	-	4	-	-
PC12. show the risks associated with the specific solar project and mitigate those	-	4	-	-
PC13. show how to support a site Feasibility Study Report using specialized software like PV*SOL®, PVsyst, etc	-	5	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	18	32	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N1207
NOS Name	Electrical feasibility study for ground mount solar power plant
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Site survey
NSQF Level	5
Credits	3
Version	2.0
Last Reviewed Date	NA
Next Review Date	27/01/2025
NSQC Clearance Date	27/01/2022

Qualification Pack

SGJ/N0402: Interpret Electrical drawing of ground mount solar plant

Description

This unit is about interpreting Electrical drawing of ground mount solar plant

Scope

The scope covers the following :

- Interpretation of electrical drawings for DC and AC side of electrical component of ground mount solar power plant

Elements and Performance Criteria

Interpretation of electrical drawings for DC and AC side of ground mount solar power plant

To be competent, the user/individual on the job must be able to:

- PC1.** discuss how to interpret the electrical single line diagram of DC side of the interconnection
- PC2.** interpret the electrical line diagram of ACDB, LT Panel, earthing layout
- PC3.** discuss how to installation and interconnection of the Inverter, DC-Disconnect, String monitoring box (SMB)
- PC4.** oversee installation of the DC and AC Energy Meter
- PC5.** discuss how to ensure the quality and workmanship of Cable laying and other electrical installation
- PC6.** interpretate the electrical single diagram of AC side of the plant
- PC7.** prepare bill of quantity and internal wiring diagram of major electrical component
- PC8.** check activities at pre commissioning and commissioning of solar Plant in both DC and AC side
- PC9.** discuss how to prepare BOQ with the with as built drawing of electrical interconnection and internal wiring diagram of major electrical component
- PC10.** discuss the process of Pre commissioning and commissioning activities of solar Plant in both DC and AC side.
- PC11.** show how to interpret the electrical drawing of DC side and AC side
- PC12.** show how to read the electrical single line diagram of HT interconnection
- PC13.** show how to read the wiring diagram of LT Panel, ACDB, SMB and other electrical equipment
- PC14.** show how to prepare the electrical BOQ from Single line diagram

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** solar cells/ modules / module technologies
- KU2.** mechanical and electrical features necessary for the long life of the PV Power Plant under a wide range of operating conditions



Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** present information in a logical and organized way
- GS2.** interpret manuals on health and safety instructions, memos, other company documents
- GS3.** read and interpret data from various sources
- GS4.** take decision with systematic course of actions and/or response
- GS5.** plan to utilise time and equipment effectively
- GS6.** use reasoning skills to identify and resolve basic problems

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Interpretation of electrical drawings for DC and AC side of ground mount solar power plant</i>	30	20	-	-
PC1. discuss how to interpret the electrical single line diagram of DC side of the interconnection	3	-	-	-
PC2. interpret the electrical line diagram of ACDB, LT Panel, earthing layout	3	-	-	-
PC3. discuss how to installation and interconnection of the Inverter, DC-Disconnect, String monitoring box (SMB)	3	-	-	-
PC4. oversee installation of the DC and AC Energy Meter	3	-	-	-
PC5. discuss how to ensure the quality and workmanship of Cable laying and other electrical installation	3	-	-	-
PC6. interpretate the electrical single diagram of AC side of the plant	3	-	-	-
PC7. prepare bill of quantity and internal wiring diagram of major electrical component	3	-	-	-
PC8. check activities at pre commissioning and commissioning of solar Plant in both DC and AC side	3	-	-	-
PC9. discuss how to prepare BOQ with the with as built drawing of electrical interconnection and internal wiring diagram of major electrical component	3	-	-	-
PC10. discuss the process of Pre commissioning and commissioning activities of solar Plant in both DC and AC side.	3	-	-	-
PC11. show how to interpret the electrical drawing of DC side and AC side	-	5	-	-
PC12. show how to read the electrical single line diagram of HT interconnection	-	5	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. show how to read the wiring diagram of LT Panel, ACDB, SMB and other electrical equipment	-	5	-	-
PC14. show how to prepare the electrical BOQ from Single line diagram	-	5	-	-
NOS Total	30	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0402
NOS Name	Interpret Electrical drawing of ground mount solar plant
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Electrician, General
NSQF Level	5
Credits	2
Version	2.0
Last Reviewed Date	NA
Next Review Date	27/01/2025
NSQC Clearance Date	27/01/2022

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SGJ/N0401: Manage AC and DC side of ground mount solar PV power plant

Description

This unit is about managing AC and DC side of ground mount solar PV power plant

Scope

The scope covers the following :

- Managing electrical installation in Ground mount solar plant

Elements and Performance Criteria

Managing electrical installation in Ground mount solar plant

To be competent, the user/individual on the job must be able to:

- PC1.** discuss how to identify manufacturers and suppliers through various sources
- PC2.** ensure the electrical installation tools shall be available on site
- PC3.** ensure all the electrical tools utilised for system installation
- PC4.** ensure testing and commissioning of equipment in DC and AC side of the plant
- PC5.** discuss how to commissioning of transformer and other HT equipment
- PC6.** ensure the electrical safety at site
- PC7.** discuss how to get CEIG/Electrical inspector inspection done at site
- PC8.** show how to prepare electrical installation plan
- PC9.** show how to manage the inventory at site
- PC10.** show how to ensure quality system installation
- PC11.** show how to manage vendor or supplier for material procurement
- PC12.** show how to prepare a vendor list of potential suppliers
- PC13.** show how to negotiate with supplier

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** power scenario of the country
- KU2.** solar energy and its landscape in the country
- KU3.** various types of land purchase and transfer procedures
- KU4.** various stages of solar power project and project requirements in terms of material, manpower and equipment at different phases
- KU5.** various project initiation and management metrics and milestones
- KU6.** various safety and crisis management protocols and PPEs at site

Generic Skills (GS)



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User/individual on the job needs to know how to:

- GS1.** how to interpret data from various sources
- GS2.** ensure organization rule-based decision making process is being followed
- GS3.** generate solutions to specific problems for a wide range of activities

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Managing electrical installation in Ground mount solar plant</i>	21	29	-	-
PC1. discuss how to identify manufacturers and suppliers through various sources	3	-	-	-
PC2. ensure the electrical installation tools shall be available on site	3	-	-	-
PC3. ensure all the electrical tools utilised for system installation	3	-	-	-
PC4. ensure testing and commissioning of equipment in DC and AC side of the plant	3	-	-	-
PC5. discuss how to commissioning of transformer and other HT equipment	3	-	-	-
PC6. ensure the electrical safety at site	3	-	-	-
PC7. discuss how to get CEIG/Electrical inspector inspection done at site	3	-	-	-
PC8. show how to prepare electrical installation plan	-	5	-	-
PC9. show how to manage the inventory at site	-	5	-	-
PC10. show how to ensure quality system installation	-	5	-	-
PC11. show how to manage vendor or supplier for material procurement	-	5	-	-
PC12. show how to prepare a vendor list of potential suppliers	-	5	-	-
PC13. show how to negotiate with supplier	-	4	-	-
NOS Total	21	29	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0401
NOS Name	Manage AC and DC side of ground mount solar PV power plant
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Management
NSQF Level	5
Credits	3
Version	2.0
Last Reviewed Date	NA
Next Review Date	27/01/2025
NSQC Clearance Date	27/01/2022

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification assessment, every trainee should score a minimum of 70% of % aggregate marks to successfully clear the assessment.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N0111.Set up New Venture for Solar Photovoltaic System	67	33	0	0	100	25
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	0	0	50	13
SGJ/N0106.Maintain Personal Health & Safety at project site	21	29	0	0	50	13
Total	108	92	-	-	200	51

Elective: 1 Rooftop

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N0108.Carry out market research and prepare a cost estimate for a Rooftop Solar Photovoltaic plant	29	21	0	0	50	12
SGJ/N0109.Prepare a site feasibility study report	60	40	0	0	100	12
SGJ/N0148.Manage Installation, Operation& Maintenance of Solar Rooftop Power Plant	54	46	0	0	100	12

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N0110.Manage Solar PV Project Cycle	29	21	0	0	50	13
Total	172	128	-	-	300	49

Elective: 2 Pumping

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N0143.Develop Solar PV Pumping Business	29	21	0	0	50	12
SGJ/N1206.Prepare a site feasibility of Solar PV water pump	30	20	0	0	50	12
SGJ/N0405.Manage Installation, operation and maintenance of Solar water pumps	52	48	0	0	100	12
SGJ/N0404.Manage Solar PV water pumping Project lifecycle	29	21	0	0	50	13
Total	140	110	-	-	250	49

Elective: 3 Ground mounted- Civil

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N0125.Survey site for installation of ground mount solar PV power Plant	30	20	0	0	50	25
SGJ/N0301.Interpret civil and structural design of solar PV power plant	32	18	0	0	50	12

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N0403.Manage civil foundation and mounting structure for installation of solar PV power plant	32	18	0	0	50	12
Total	94	56	-	-	150	49

Elective: 4 Ground mounted-Electrical

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N1207.Electrical feasibility study for ground mount solar power plant	18	32	0	0	50	25
SGJ/N0402.Interpret Electrical drawing of ground mount solar plant	30	20	0	0	50	12
SGJ/N0401.Manage AC and DC side of ground mount solar PV power plant	21	29	0	0	50	12
Total	69	81	-	-	150	49

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.